Lecture 9
More on Linux and Bash

Comp 206 – Introduction to Software Systems – Fall 2026

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Reminders

- Class Test 1: Tuesday Oct 6, class time (14:35-15:55)
  - Exam is 75 minutes (1h15)
  - Letter-sized (8.5” by 11”) handwritten crib sheet, single or double-sided
- MyCourses:
  - Practice Midterm 1 from Winter 2026
  - Practice problems written by TA
  - Feel free to discuss both of these on Ed!
- Homework:
  - Reminder that the hard deadline for homework 1 to 4 is Oct 6, the day of Class Test 1!
- Next quiz: Thursday Oct 8

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Remaining tutorials before Class Test 1

- Tutorials:
  - Tutorial 3: Lab C Intro to Bash
    - Wednesday Sep 30 ENGTR 3120, 16:00-17:00 (TA Siam)
      - (This used to be the Monday 8:30 tutorial, see Ed post)
  - Tutorial 4: Solving a TA Bash practice problem
    - Friday Oct 2 ENGTR 2110, 13:35-14:25 (TA Guru)
    - Monday Oct 5 ENGTR 2120, 10:35-11:25 (TA Michael)
    - Monday Oct 5 ENGTR 1090, 14:35-15:25 (TA Abidur)
    - (Next tutorial after this: Friday Oct 16, Intro to C)

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Remaining study sessions before Class Test 1

- Study sessions (Trottier 3rd floor):
  - Thursday Oct 1st 10:30-11:30
  - Friday Oct 2nd 14:15-15:15
  - Tuesday Oct 6th 13:00-14:00
- Considering working on your assignments and studying for the classes there! Two TAs will be walking around to help you!

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Remaining OH before Class Test 1

- My drop-in OH: Wednesday Sep 30 14:30-**15:30**
- My appointment OH: Some times remaining on Monday Oct 5
- TA Mentorship/OH appointments: Many times still available every day until the exam.

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More on Linux and Bash

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Demo

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Quiet

- In case it's undesirable to see the output of a command run inside a script, there are a few ways to suppress a command's output:
  - Some commands come with flags to suppress their output, e.g. `grep -q` (the q stands for "quiet").
  - Redirect the output into `/dev/null` -- this is a special file that doesn't actually store any data and just ignores anything we try to write to it
    - Example: `ls > /dev/null 2> /dev/null`
      - Redirects stdout and stderr to /dev/null
    - Equivalently: `ls > /dev/null 2>&1`
      - Redirects stdout to /dev/null, and redirects stderr to stdout

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Environment variables

- Environment variables contain information about your login session (The time between login and logout)
  - Examples:
    - $HOME
    - $PATH
    - $PS1
- Just like any other variables, they can be edited
  - Often, we want *child processes* (e.g. scripts you run from your Shell) to inherit these variables: the `export` keyword does this
    - `export $PATH="$PATH:~/my_dir"`
      - Adds `~/my_dir` to your `$PATH`
  - Variable modifications are only visible inside that particular session

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System scripts

- Used by the shell for configuration purposes
- Examples:
  - ~/.bash_profile: Executed when you login a Bash shell
  - ~/bashrc: Executed whenever an interactive non-login Bash shell is opened
- Can use this to set environment variables every time you login!
- Example:
  - Add `export PS1="[\t]\u@\h:\w\$` to `~/.bash_profile`

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Alias

- The alias command
  - You give an alternate name to commands
  - Syntax:
    - `alias NEWNAME=OLDEXPRESSION`
  - Example:
    - `alias dir='ls -l -a'`
    - What if you did `alias dir='ls -l -a'` instead?

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Tar

- An archive is a collection of files combined into one file.
    - Being one file, archives are easier to manipulate (move, store, copy, backup, etc).
    - Archives are often compressed, so they require less space.
- The two most common archive tools used on Unix systems is tar and gzip (gunzip).
- Tar allows the manipulation (creation, extraction) of archive files.
    - Convention:
        - A file ending with the .tar extension is a tar archive file.
        - A file ending with the .tgz extension is a compressed (gzipped) tar archive file.
- Switches:
    - -c : create a new tar archive
    - -r : update the tar archive
    - -x : extract from the tar archive
    - -f: specifies the archive file name.
    - -v: activates verbose mode, which means the tar command will output lots of information.
    - -z: allows you to compress the archive (the archive is compressed/decompressed using gzip).

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Diff

- Shows the difference between two files
- Syntax:
  - `diff [options] file1 file2`

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ln: hard and symbolic links

- `ln` and `ln -s`
- The ln command can be used to create links to files and folders.
  - Hard link: ln /path/file link_name
  - Soft link: ln –s /path/file link_name
- When creating a hard link, you are simply giving another name to a file (it shows up separately on an ls – a direct pointer in directory)
  - The link will point to the same physical space on the disk.
  - A file can only be deleted once all its hard link are deleted.
- When creating a symbolic link (using ln -s), a new file is created (an indirect pointer in directory)
  - The new file automatically redirects to the target file.
  - Symbolic links can be created across volumes (or disks).
  - Deleting a symbolic link does not affect the target file.
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Additional credits

- Some content from this lecture is based on:
  - Past Comp 206 slides/notes from Prof. Vybihal

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